SUMMARY OF SAFETY AND EFFECTIVENESS DATA

I. GENERAL INFORMATION

Device Generic Name:	Cutting/scoring percutaneous transluminal coronary angioplasty catheter
Device Trade Name:	AngioSculpt® Scoring Balloon Catheter
Applicant's Name and Address:	AngioScore, Inc. 5055 Brandin Court Fremont, CA 94538
Premarket Approval Application (PMA) Number:	P050018
Date of Panel Recommendation:	None
Date of Notice of Approval To Applicant:	January 8, 2007

II. INDICATIONS FOR USE

The AngioSculpt® Scoring Balloon Catheter is indicated for the treatment of hemodynamically significant coronary artery stenosis, including in-stent restenosis and complex type C lesions, for the purpose of improving myocardial perfusion.

III. CONTRAINDICATIONS

The AngioSculpt catheter should not be used in:

- coronary artery lesions unsuitable for treatment by percutaneous revascularization
- coronary artery spasm in the absence of a significant stenosis.

IV. WARNINGS AND PRECAUTIONS

Please refer to the device labeling for a list of warnings and precautions.

V. DEVICE DESCRIPTION

The AngioSculpt® Scoring Balloon Catheter is a standard PTCA catheter with a scoring balloon near the distal tip. The balloon is designed to expand to a specified diameter and length at a specified pressure. The distal end of the catheter has a conventional nylon-blend balloon and a nitinol scoring element with three spiral struts that wrap around the balloon. The struts create focal concentrations of dilating force, which minimizes balloon slippage and assists in the luminal expansion of stenotic arteries. Conventional radiopaque markers aid in positioning the balloon in the stenosis.

The product is offered in over-the-wire (OTW) and easy exchange (EX) delivery platforms and is available in balloon diameters of 2.0 – 3.5 mm in 0.5 mm increments, and in scoring balloon lengths of 10, 15 and 20 mm. The catheter length is approximately 137 cm and is compatible with 0.014-inch guide wires and 6F guide catheters. The catheter is supplied sterile and is intended for a single use.

VI. ALTERNATIVE PRACTICE AND PROCEDURES

Alternative treatments for coronary artery disease include medical therapy, atherectomy, laser, stent, endarterectomy, bypass graft (CABG) surgery and other commercially available PTCA catheters.

VII. MARKETING HISTORY

AngioScore, Inc. obtained CE Mark certification in September 2004 for its Interventional Cardiology Devices and Accessories, including the AngioSculpt catheter. The device has been commercialized in the United Kingdom, Italy, Spain, and Israel.

The AngioSculpt catheter has not been withdrawn from any country for reasons relating to device safety and effectiveness.

VIII. SUMMARY OF NON-CLINICAL STUDIES

The AngioSculpt catheter has been evaluated in accordance with FDA's Guidance for the Submission of Research and Marketing Applications for Interventional Cardiology Devices, consultations with FDA, and relevant industry standards.

A. Biocompatibility Testing

The materials used in the AngioSculpt catheter have a proven history of safe use in commercially available interventional cardiology devices. **Table 1** summarizes confirmatory, GLP biocompatibility testing that was conducted on representative samples of finished, sterilized AngioSculpt catheters. The product is categorized as an "external communicating device in contact with circulating blood with limited exposure time".

All biocompatibility testing was conducted in accordance with:

- FDA's Guidance for the Submission of Research and Marketing Applications for Interventional Cardiology Devices; May 1995.
- ISO 10993-1: "Biological evaluation of medical devices – Part 1: Evaluation and testing."
- FDA's Use of International Standard ISO-10993, 'Biological Evaluation of Medical Devices Part 1: Evaluation and Testing'; 1995 Bluebook Memorandum.

Table 1: Biocompatibility Test Summary

	Test Performed	Results
Cytotoxicity	ISO 10993-5: *In Vitro* Cytotoxicity (L929 MEM Elution Method)	**Pass; non-cytotoxic**
Sensitization	ISO 10993-10: Guinea Pig Maximization Method	**Pass; non-sensitizing**
Intracutaneous Reactivity	ISO 10993-10: Irritation (Rabbit)	**Pass; non-irritating**
Acute Systemic Toxicity	ISO 10993-11: Acute Systemic Injection (Mouse)	**Pass; no evidence of systemic toxicity**
Materials-Mediated Pyrogenicity	ISO 10993-11 & USP <26>: Material Mediated Pyrogenicity (Rabbit)	**Pass; non-pyrogenic (material-mediated)**
Hemocompatibility		
Hemolysis / Hematology	ISO 10993-4: NIH Direct Contact Method	**Pass; non-hemolytic**
Thrombogenicity	ISO 10993-4: *In Vitro* Partial Thromboplastin Time (PTT)	**Pass; same as negative control**
	ISO 10993-4: *In Vivo* Thrombogenicity (Beagle Dogs; 4-Hours)	**Slightly higher thrombogenicity potential than control***
Complement Activation	ISO 10993-4: *In Vitro* C3a & SC5b-9	**Pass; comparable to currently marketed stent device**

*Maximum thrombus scores in the AngioSculpt catheter and the control, a currently marketed device, were the same (grade 1 out of 4).

The test results confirm that the AngioSculpt finished device materials are not likely to provoke an adverse biological response when used as intended in direct blood contact for a limited duration.

B. *In Vitro* Engineering Testing

In vitro engineering bench testing of the AngioSculpt catheter was conducted to verify that the design outputs meet the design requirements and to confirm the safety and performance of the product from a non-clinical testing perspective.

Testing was performed, as applicable, in compliance with:

- FDA's Guidance for the Submission of Research and Marketing Applications for Interventional Cardiology Devices: PTCA Catheters, Atherectomy Catheters, Lasers and Intravascular Stents; 1995 draft.
- FDA's Guidance: Non-Clinical Tests and Recommended Labeling for Intravascular Stents and Associated Delivery Systems - Guidance for Industry and FDA Staff; January 2005.
- Relevant industry standards.

Table 2 provides a summary of the mechanical performance bench testing conducted on the AngioSculpt catheter.

Table 2: Mechanical Performance Bench Test Summary

Test	Description of Test & Test Articles	Conclusion
Balloon Rated Burst Pressure	Testing was performed on AngioSculpt catheters (sizes representative of the available product) to establish the rated burst pressure (RPB). The results demonstrated with 95% confidence that at least 99.9% of the balloons will not experience loss of integrity at or below the labeled RBP.	Pass
Balloon Inflation and Deflation	Testing was performed on AngioSculpt catheters to verify that the balloon inflates and deflates within a specified time. All samples tested met specifications with 95% confidence and 90% reliability.	Pass
Balloon and Scoring Element Fatigue	AngioSculpt catheters were subjected to 20 consecutive inflation cycles to the RBP. The results demonstrated with 95% confidence that at least 90% of the catheters can sustain 20 consecutive inflation cycles to the RBP without failure. A Finite Element Analysis was performed to determine the stress/strain levels of the spiral wires. Results of the FEA confirm that a mechanical failure is not likely to occur as a result of over-stress.	Pass
Diameter vs. Inflation Pressure (Balloon Compliance)	Testing was performed on AngioSculpt catheters to determine how the diameter of a balloon varies with applied inflation pressure. The results were used to establish the labeled compliance values.	Pass
Catheter Body Integrity	AngioSculpt catheters were inflated above RBP when tested for balloon burst. No catheter body failures were observed in any of the cases.	Pass
Bond Strength of Catheter	Testing was performed on AngioSculpt catheters to ensure that all bond joints on the catheter would meet tensile strength specifications. All bonds tested met specifications with 95% confidence and 95% reliability.	Pass
Scoring Element Bond Strength	Testing was performed on AngioSculpt catheters to ensure that scoring element to catheter bonds would meet tensile strength specifications. All bonds tested met specifications with 95% confidence and 95% reliability.	Pass
Tip Pull Strength	Testing was performed on AngioSculpt catheters to demonstrate that the catheter soft tip bond would meet tensile strength specifications. All bonds tested met specifications with 95% confidence and 95% reliability.	Pass
Crossing Profile	Testing was performed on AngioSculpt catheters to determine the scoring balloon profile. All samples tested met specifications with at least 95% confidence and 90% reliability.	Pass

Test	Description of Test & Test Articles	Conclusion
Trackability and Pushability	Testing was performed on AngioSculpt catheters to demonstrate that the AngioSculpt catheter could track simulated anatomical curves with a reasonable push force. Results demonstrated that the average peak tracking forces for the AngioSculpt catheter were lower than for other similar commercially-available devices.	Pass
Over-the-arch torque strength	Testing was performed on AngioSculpt catheters to assess the ability of the AngioSculpt catheter to withstand over-the-arch torque when the distal tip is not free to rotate. All samples tested met specifications with at least 90% confidence and 90% reliability.	Pass
Freedom from Interference	Testing was performed on AngioSculpt catheters to verify that the scoring element would not interfere with implanted stents, using a bench model of under-deployed stents in a simulated tortuous anatomy. Results demonstrated freedom from interference with under-deployed stents with 95% confidence and 95% reliability.	Pass
Secure Edges	Testing performed on AngioSculpt catheters demonstrated that the scoring element is secured to the catheter without free edges that might interfere with a guiding catheter during retraction. No interference with the guide catheter was observed in mechanical performance, *in vivo* simulated use (human cadaver and porcine studies) and clinical studies.	Pass
Radiopacity	Radiopacity of the AngioSculpt catheter Pt/Ir markers verified during cadaver, animal, and clinical studies showed that the catheter is visible when used as intended in clinical applications.	Pass
Catheter Effect on Coating Integrity of Drug Eluting Stents	Testing was performed to determine the effect of the AngioSculpt catheter on the coating integrity of drug-eluting stents. No significant differences were observed in the observed defect rate compared to the control group (untreated stents).	Pass
Catheter Effect on Structural Integrity of Implantable Stents	Testing was performed to determine the effect of the AngioSculpt catheter on the integrity of stent struts. Results demonstrated that use of the AngioSculpt device resulted in no damage to stent struts or structure, with 95% confidence and at least 99% reliability.	Pass
Catheter Preparation, Visual and Dimensional Verification	AngioSculpt catheters were dimensionally inspected, visually inspected, and subjected to inflation and catheter body integrity tests. All units met specifications.	Pass

C. Shelf Life, Packaging, and Sterilization Testing

Package testing was completed to evaluate the effects of distribution, handling and storage on the AngioSculpt catheter device and package configuration.

The package and shelf life testing complied with requirements in:

- ISO 11607:2003E "Packaging for terminally sterilized medical devices."

- ASTM F1980-02 "Standard Guide for Accelerated Aging of Sterile Medical Device Packages."
- ASTM F2097-01 "Standard Guide for Design and Evaluation of Primary Packaging for Medical Products."

Results from testing conducted on finished product which was aged for at least one year demonstrate the robustness of the AngioSculpt product and packaging, and support a one (1) year shelf life.

The AngioSculpt catheter is sterilized using a validated gaseous ethylene oxide (EO) sterilization process that provides a sterility assurance level (SAL) of at least 10^{-6}. The device is labeled for single use only. EO residuals have been demonstrated through testing to be within acceptable limits, as described in ISO 10993-7: 1995 (R) 2001, "Biological Evaluation of Medical Devices – Part 7: Ethylene Oxide Sterilization Residuals."

D. *In Vivo* Simulated Use Testing

Testing in human cadaver and animal models demonstrated that the AngioSculpt catheter was appropriate for human clinical evaluation. A summary of *in vivo* simulated use testing conducted on the AngioSculpt catheter is provided in **Table 3**.

Table 3: Simulated Use Tests

Test	Summary
Human Cadaver Testing	Performance of the AngioSculpt catheter was evaluated in severely diseased amputated human legs. All lesions were successfully treated with resultant residual diameter stenosis of <10%. Pre- and post-angiograms and IVUS images revealed no significant dissections and no perforations at the site of AngioSculpt catheter treatment. Most lesions yielded at low device inflation pressures (4-6 atmospheres). Gross and histopathology confirmed the absence of vessel perforation or other unexpected injury.
Porcine Native Vessel Deliverability Studies	The AngioSculpt catheter was tested in normal porcine coronary arteries to evaluate device deliverability and safety. Using standard interventional techniques the AngioSculpt catheter was deployed at several sites in multiple coronary arteries of several animals. The device was successfully delivered to proximal, mid and distal locations in each coronary artery attempted. Device integrity was maintained throughout the procedures. Post-procedural coronary angiography revealed no evidence of dissection, perforation, spasm or distal embolization for all locations treated.

Test	Summary
Porcine ISR Studies	The AngioSculpt catheter was tested in acute and chronic (~ 4 week) porcine coronary artery balloon overstretch/stent injury models to evaluate device performance in an accepted porcine coronary artery model of in-stent restenosis. Acceptable residual diameter stenosis was demonstrated by angiography immediately following AngioSculpt treatment. Post-treatment IVUS demonstrated significant improvement in luminal dimensions with "scoring" marks visible consistent with the geometry of the nitinol wires of the device. No dissections or vessel perforations were revealed on coronary angiography or intravascular ultrasound. No animals suffered any complications, hemodynamic instability or unexpected death during the procedure. No device "slippage" was observed during any AngioScore catheter treatment. Histopathologic evaluation of treated segments from animals sacrificed immediately after the treatment demonstrated clear-cut neointimal "scoring" marks at the presumed sites of nitinol wire impact with no evidence of vessel wall perforation. Approximately 28 days post AngioSculpt treatment, the angiographic data showed no signs of detectable dissection, vessel perforation, distal embolization, thrombus formation, aneurysm formation, filling defects, or other adverse cardiac events. The 28-day histology showed low injury scores and mild to moderate intimal response to AngioSculpt catheter treatment; the results were similar to that of balloon angioplasty or stenting.

IX. POTENTIAL ADVERSE EVENTS

A. Potential Adverse Events

Adverse events which may be associated with balloon angioplasty, including the use of a balloon with a scoring element, may include:

- Death
- Heart Attack (acute myocardial infarction)
- Total occlusion of the treated coronary artery
- Coronary artery dissection, perforation, rupture, or injury
- Pericardial tamponade
- No/slow reflow of treated vessel
- Emergency coronary artery bypass (CABG)
- Emergency percutaneous coronary intervention
- CVA/stroke
- Pseudoaneurysm
- Restenosis of the dilated vessel
- Unstable angina
- Thromboembolism or retained device components
- Irregular heart rhythm (arrhythmias, including life-threatening ventricular arrhythmias)
- Severe low (hypotension)/high (hypertension) blood pressure
- Coronary artery spasm
- Hemorrhage or hematoma
- Need for blood transfusion
- Surgical repair of vascular access site
- Creation of a pathway for blood flow between the artery and the vein in the groin (arteriovenous fistula)
- Drug reactions, allergic reactions to x-ray dye (contrast medium)
- Infection

B. Observed Adverse Events

A total of 200 patients were enrolled in the AngioSculpt catheter pivotal IDE clinical trial, a prospective, multi-center, non-randomized, single-arm study. The primary objective of the study was to evaluate safety and efficacy of the AngioSculpt catheter in a wide range of coronary artery lesions in both native vessels and following in-stent restenosis (ISR).

Table 4 provides a summary of the Major Adverse Cardiac Events (MACE) observed in the AngioSculpt catheter multi-center IDE study, as determined by the Clinical Events Committee (CEC). MACE was defined as death, Q wave or non-Q wave myocardial infarction (MI) or target lesion revascularization (TLR).

Table 4: Summary of Principal Adverse Event Rates Observed in the IDE Study

	Aggregate	95% C.I.
In-Hospital Complications	**% of patients (number of events)**	
MACE (Death, MI, TLR)	1.5% (3/200)	(0.3%, 4.3%)
Death	0.0% (0/200)	(0.0%, 1.8%)
MI (Q or Non-Q wave)	1.5% (3/200)	(0.3%, 4.3%)
Q wave MI	0.5% (1/200)	(0.0%, 2.8%)
Non-Q wave MI*	1.0% (2/200)	(0.1%, 3.6%)
Target Lesions Revascularization (TLR)	0.0% (0/200)	(0.0%, 1.8%)
TLR PTCA	0.0% (0/200)	(0.0%, 1.8%)
TLR CABG	0.0% (0/200)	(0.0%, 1.8%)
Target Vessel Revascularization(TVR)	0.0% (0/200)	(0.0%, 1.8%)
TVR PTCA	0.0% (0/200)	(0.0%, 1.8%)
TVR CABG	0.0% (0/200)	(0.0%, 1.8%)
Thrombosis	0.5% (1/200)	(0.0%, 2.8%)
Acute Thrombosis	0.5% (1/200)	(0.0%, 2.8%)
Confirmed	0.0% (0/200)	(0.0%, 1.8%)
Presumed	0.5% (1/200)	(0.0%, 2.8%)
Sub-Acute Thrombosis	0.0% (0/200)	(0.0%, 1.8%)
Confirmed	0.0% (0/200)	(0.0%, 1.8%)
Presumed	0.0% (0/200)	(0.0%, 1.8%)
Out-of-hospital Complications		
MACE (Death, MI, TLR)	1.0% (2/200)	(0.1%, 3.6%)
Death	0.0% (0/200)	(0.0%, 1.8%)
MI (Q or Non-Q)	1.0% (2/200)	(0.1%, 3.6%)
Q wave MI	1.0% (2/200)	(0.1%, 3.6%)
Non-Q wave MI	0.0% (0/200)	(0.0%, 1.8%)
TLR	1.0% (2/200)	(0.1%, 3.6%)
TLR PTCA	0.5% (1/200)	(0.0%, 2.8%)
TLR CABG	0.5% (1/200)	(0.0%, 2.8%)
Target Vessel Revascularization(TVR)	1.0% (2/200)	(0.1%, 3.6%)
TVR PTCA	0.5% (1/200)	(0.0%, 2.8%)
TVR CABG	0.5% (1/200)	(0.0%, 2.8%)
Sub-Acute Thrombosis	1.0% (2/200)	(0.1%, 3.6%)
Confirmed	1.0% (2/200)	(0.1%, 3.6%)
Presumed	0.0% (0/200)	(0.0%, 1.8%)
Cumulative Complications		
MACE (Death, MI, TLR)	2.5% (5/200)	(0.8%, 5.7%)
Death	0.0% (0/200)	(0.0%, 1.8%)
MI (Q or Non-Q)	2.5% (5/200)	(0.8%, 5.7%)
Q wave MI	1.5% (3/200)	(0.3%, 4.3%)

Table 4: Summary of Principal Adverse Event Rates Observed in the IDE Study		
	Aggregate	95% C.I.
Non-Q wave MI	1.0% (2/200)	(0.1%, 3.6%)
TLR	1.0% (2/200)	(0.1%, 3.6%)
TLR PTCA	0.5% (1/200)	(0.0%, 2.8%)
TLR CABG	0.5% (1/200)	(0.0%, 2.8%)
Target Vessel Revascularization (TVR)	1.0% (2/200)	(0.1%, 3.6%)
TVR PTCA	0.5% (1/200)	(0.0%, 2.8%)
TVR CABG	0.5% (1/200)	(0.0%, 2.8%)
Thrombosis	1.5% (3/200)	(0.3%, 4.3%)
Confirmed	1.0% (2/200)	(0.1%, 3.6%)
Presumed	0.5% (1/200)	(0.0%, 2.8%)

* Non-Q wave MI is defined as elevation of the CPK to >3x the upper limit of normal (associated with an abnormally elevated Troponin-I) without the development of new pathologic Q-waves.

In addition to MACE, a total of 23 non-MACE cardiovascular events and 37 non-cardiovascular events were reported by the clinical sites. Among these events, the principal cardiovascular events were classified as unstable angina (3%), coronary artery injury (dissection, perforation, or rupture, 1.5%), hypotension (1.5%), and excessive blood loss/hemorrhage requiring transfusion or hematoma (1.5%). These events have been associated with use of other PTCA catheters.

C. Observed Device Malfunctions

In total, there were 16 reported failures to cross the lesion with the AngioSculpt catheter. Fifteen of these failures were adjudicated as device-related. There was one failure to cross the lesion which was adjudicated by the Clinical Events Committee as procedure-related but unrelated to the study device. All sixteen lesions were subsequently successfully pre-dilated with a commercially available balloon followed by stent placement.

There were two reported cases of device malfunction due to loss of balloon pressure below rated burst pressure. These device malfunctions were unassociated with any adverse event.

X. SUMMARY OF CLINICAL STUDIES

The safety and performance of the AngioSculpt catheter has been evaluated through domestic and international clinical investigations. The pivotal clinical study was a multi-center clinical investigation that enrolled two-hundred (200) patients at nine (9) centers in the United States. Prior to the pivotal study, AngioScore sponsored clinical investigations outside of the United States (OUS) involving two (2) centers and forty-five (45) patients. Results from these studies demonstrate that the AngioSculpt catheter is safe with an acceptably low rate of complications and effective in a broad range of coronary artery lesions consistent with its intended use.

A. IDE Clinical Trial

Objective: To evaluate the safety and efficacy of the AngioSculpt catheter in a wide range of coronary artery lesions in both native vessels and following in-stent restenosis (ISR).

Conclusion: Results of the pivotal U.S. multi-center clinical investigation demonstrate that the AngioSculpt Scoring Balloon Catheter is safe and effective for use in PTCA procedures. The procedural success rate, clinical success rate, and MACE rate and type observed in this investigation met the objective performance criteria developed from published scientific literature for other PTCA balloon catheters. There was no increase in the observed rate of perforations or dissections. The investigation results also demonstrate that the AngioSculpt Scoring Balloon Catheter provides a stable, non-slip platform that can be safely used to treat *de novo* coronary lesions (including complex type C lesions) and ISR lesions.

Design: A prospective, multi-center, non-randomized, single arm investigational study was conducted in the United States at nine (9) sites enrolling two-hundred (200) patients. Clinical sites followed an investigational plan and protocol approved under an FDA-approved IDE application. The investigation was designed to evaluate the following primary performance and safety objectives:

- To compare a procedural success rate to objective performance criteria for percutaneous coronary intervention derived from contemporary published literature.
- To compare a clinical success rate to objective performance criteria for percutaneous coronary interventions derived from contemporary published literature.

Endpoints: The primary effectiveness endpoint was procedural success, defined as ≤50% final diameter stenosis in at least one of the AngioSculpt-attempted lesions following completion of the interventional procedure, including adjunctive stenting when used, without death, Q wave or non-Q wave MI, or emergency CABG surgery during the hospital stay. The primary safety endpoint was clinical success rate, defined as freedom from MACE [death, Q wave or non-Q wave MI, or target lesion revascularization (TLR)] at 14-day follow-up.

Eligibility Criteria: The AngioSculpt U.S. clinical investigation study population included patients with single or multiple vessel coronary artery disease who were scheduled to undergo percutaneous coronary intervention because of symptoms of stable or unstable angina. No more than two lesions were to be treated during the index procedure. Key patient selection criteria were:

- At least 18 years of age and able to give informed consent
- Stenotic lesions in native coronary arteries, including in-stent restenosis
- Lesions with a reference vessel diameter (RVD) between 2.0 mm and 4.0 mm.
- Target lesions meeting the following criteria were cause for exclusion:
  - Totally obstructed coronary arteries (TIMI 0 or 1 flow)

- Lesions longer than 30 mm
- Target lesion in a severe angulation (> 60 degrees)
- Target lesion distal to a freshly deployed stent
- Target lesion within a previously-placed polymer-coated stent (drug or non-drug eluting), unless complete tissue coverage of the stent was verified with IVUS
- Target lesion demonstrating severe dissection prior to planned deployment of the AngioSculpt device
- Visible thrombus at the target lesion
- Unprotected left main artery
- Severe calcification
- Lesions in surgical conduits

Baseline Characteristics: Following informed consent, 200 patients (age 63.5 ± 11.3 years, males 72%) referred for percutaneous coronary intervention underwent treatment with the AngioSculpt catheter. Of these patients, 27% (54/200) had a history of diabetes mellitus. Unstable angina was present in 31.5% (63/200) of patients. Other relevant clinical characteristics are described in **Table 5**, and are consistent with that of typical patients in whom the AngioSculpt catheter is likely to be used.

Table 5: Baseline Patient Demographics and Clinical Characteristics

	Aggregate	Range	95% C.I.
Age	63.45 +/- 11.32 (200)	(31.44, 90.50)	(61.87, 65.03)
Male	71.5% (143/200)		(64.7%, 77.6%)
Female	28.5% (57/200)		(22.4%, 35.3%)
Race-Caucasian	84.5% (169/200)		(78.7%, 89.2%)
Race-African American	8.0% (16/200)		(4.6%, 12.7%)
Race-Hispanic	3.5% (7/200)		(1.4%, 7.1%)
Race-Asian	1.5% (3/200)		(0.3%, 4.3%)
Race-Other	2.5% (5/200)		(0.8%, 5.7%)
BMI	29.70 +/- 5.66 (200)	(18.04, 53.50)	(28.91, 30.49)
Ejection Fraction: Normal (>50%)	61.5% (123/200)		(54.4%, 68.3%)
Ejection Fraction:35-50%	33.5% (67/200)		(27.0%, 40.5%)
Ejection Fraction:<35%	0.5% (1/200)		(0.0%, 2.8%)
Stable Angina	53.5% (107/200)		(46.3%, 60.6%)
Unstable Angina	31.5% (63/200)		(25.1%, 38.4%)
History of Diabetes	27.0% (54/200)		(21.0%, 33.7%)
Insulin Dependent Diabetes	7.5% (15/200)		(4.3%, 12.1%)
Non-insulin Dependent Diabetes	19.5% (39/200)		(14.2%, 25.7%)
History of PTCA	57.0% (114/200)		(49.8%, 64.0%)
History of CABG	15.5% (31/200)		(10.8%, 21.3%)

Table 5: Baseline Patient Demographics and Clinical Characteristics			
	Aggregate	**Range**	**95% C.I.**
History of CVA/TIA	9.0% (18/200)		(5.4%, 13.9%)
History of MI	26.0% (52/200)		(20.1%, 32.7%)
History of Smoking	58.0% (116/200)		(50.8%, 64.9%)
History of CHF	6.0% (12/200)		(3.1%, 10.2%)
History of Hypertension Requiring Medication	78.0% (156/200)		(71.6%, 83.5%)
History of Hyperlipidemia Requiring Medication	85.0% (170/200)		(79.3%, 89.6%)
History of Peripheral Vascular Disease	8.0% (16/200)		(4.6%, 12.7%)
History of Renal Insufficiency	9.0% (18/200)		(5.4%, 13.9%)
Family History of CAD	46.5% (93/200)		(39.4%, 53.7%)
Patients with only 1 Lesion Treated with ASC	84.5% (169/200)		(78.7%, 89.2%)
Patients with 2 Lesions Treated with ASC	15.5% (31/200)		(10.8%, 21.3%)

- Denominators were adjusted to include all patients and unknown information for medical history.
- ASC = AngioSculpt

Baseline qualitative and quantitative angiographic characteristics are summarized in **Table 6** below. Of the 219 treated lesions, 184 (84.0%) were *de novo* and 35 (16.0%) were treated for in-stent restenosis.

Table 6: Baseline Angiographic Characteristics			
	Aggregate	**Range**	**95% C.I.**
ACC[1] Score A	4.1% (9/219)		(1.9%, 7.7%)
ACC Score B1	19.6% (43/219)		(14.6%, 25.5%)
ACC Score B2	28.3% (62/219)		(22.4%, 34.8%)
ACC Score C	47.9% (105/219)		(41.2%, 54.8%)
Lesion Length	17.79 +/- 8.94 (219)	(6.15, 55.15)	(16.60, 18.98)
Pre-Procedure RVD	2.72 +/- 0.39 (219)	(1.93, 3.91)	(2.67, 2.78)
Pre-Procedure MLD	0.78 +/- 0.31 (219)	(0.00, 1.76)	(0.74, 0.82)
Pre-Procedure %DS	71.61 +/- 10.16 (219)	(45.73, 100.00)	(70.25, 72.96)
Lesion Angulation (degrees)	15.48 +/- 17.62 (218)	(0.00, 90.00)	(13.13, 17.83)
Eccentric Lesion	26.9% (59/219)		(21.2%, 33.3%)
Bifurcation Lesion	28.8% (63/219)		(22.9%, 35.3%)
Visible Thrombus	2.7% (6/219)		(1.0%, 5.9%)

[1] ACC/AHA Lesion Class #: Smith, et al ACC/AHA Percutaneous Coronary Intervention Guidelines. JACC 2001;37:2239i-xvi

Table 6: Baseline Angiographic Characteristics			
	Aggregate	**Range**	**95% C.I.**
Lesion Location			
Proximal	35.6% (78/219)		(29.3%, 42.3%)
Mid	37.0% (81/219)		(30.6%, 43.8%)
Distal	14.6% (32/219)		(10.2%, 20.0%)
Ostial	12.8% (28/219)		(8.7%, 17.9%)
LAD	37.9% (83/219)		(31.4%, 44.7%)
RCA	37.0% (81/219)		(30.6%, 43.8%)
LCX	23.3% (51/219)		(17.9%, 29.5%)
LM	1.4% (3/219)		(0.3%, 4.0%)
SVG	0.5% (1/219)		(0.0%, 2.5%)
Lesion Calcification*	35.3% (73/207)		(28.8%, 42.2%)
Moderate	29.0% (60/207)		(22.9%, 35.7%)
Severe	6.3% (13/207)		(3.4%, 10.5%)
TIMI Flow			
TIMI Flow 0	1.4% (3/219)		(0.3%, 4.0%)
TIMI Flow 1	0.9% (2/219)		(0.1%, 3.3%)
TIMI Flow 2	5.0% (11/219)		(2.5%, 8.8%)
TIMI Flow 3	92.7% (203/219)		(88.4%, 95.8%)

RVD = reference vessel diameter (mm), MLD = minimal lumen diameter (mm), %DS = percent diameter stenosis (%), LAD = left anterior descending, RCA = right coronary artery, LCX = left circumflex, LM = left main, SVG = saphenous vein graft

*Pre-Procedure calcification was not able to be assessed for 12 lesions in 11 patients.

Methods: Prior to treatment with the AngioSculpt catheter, patients provided informed consent and underwent routine baseline evaluations. Patients were pre-medicated according to institutional protocol for procedures involving stents, a coronary angiogram of the target lesion was obtained, and a guide wire was positioned beyond the target lesion. When used to pre-dilate the lesion prior to pre-planned stenting, physicians were instructed to select a catheter that was one size smaller relative to the estimated reference vessel diameter (RVD). For example, for a RVD of 3.0 mm by visual estimation, a 2.5mm diameter device was used. When used for in-stent restenosis, catheters were selected to closely approximate the RVD (≤ 1.0 x RVD). The AngioSculpt catheter was prepped according to the IFU and standard procedures. The AngioSculpt catheter was advanced over the coronary guide wire to the target lesion. Cineangiograms were performed to confirm proper positioning prior to balloon inflation. Additional clinically indicated interventions (e.g., stent placement) were performed at the discretion of the investigator. Anticoagulation and antiplatelet medications were prescribed per institutional protocol for procedures involving stents. Post-procedure evaluations were performed until hospital discharge and at 14-21 days following the procedure. Results of the evaluations and procedure assessments, including post-procedure adverse events and patient withdrawals, were documented on study specific case report forms (CRFs).

QCA, ECG and IVUS analysis were performed by independent Core Laboratories. All suspected MACE and device failures/malfunctions were adjudicated by an independent Clinical Events Committee.

Results: Data analysis was performed on an intent-to-treat basis. The study met its primary safety and efficacy objectives as described below and summarized in **Table 7.**

All patients and all lesions had a final diameter stenosis that was ≤ 50% in at least one AngioSculpt-attempted lesion following completion of the interventional procedure. Three patients experienced in-hospital MACE (two non-Q wave MI, one Q wave MI) contributing to a procedural success rate of 98.5% (197/200). Additionally, two patients experienced MACE post-hospital discharge (one Q wave MI with TLR-PCI and one Q wave MI with TLR-CABG). This resulted in a clinical success rate of 97.5% (195/200). All AEs, including MACE, are within the expected rates for this patient population.

Table 7: Summary of Primary Endpoint Results

	Aggregate	Lower bound one-sided 95% C.I.
Procedural Success	98.5% (197/200)	96.2%
Clinical Success	97.5% (195/200)	94.8%

Procedural Success = ≤50% final diameter stenosis in at least one AngioSculpt-attempted lesion following completion of the interventional procedure, including adjunctive stenting when used, without death Q-wave or non-Q-wave MI, or emergency CABG during the hospital stay.

Clinical Success = freedom from MACE (death, Q wave or non-Q wave MI, or TLR) at 14-day follow-up.

In all 219 lesions (100%), the angiographic component of the primary efficacy endpoint (reduction of the lesion diameter stenosis to ≤ 50% at the completion of the interventional procedure) was successfully achieved. **Table 8** summarizes the angiographic results following AngioSculpt treatment and following adjunctive stenting, which was performed in 97.7% (211/216) of cases.

The post-AngioSculpt dissection rate was 13.6% with the majority of these rated as low grade (A-C) utilizing the NHLBI classification. In all treated lesions, the AngioSculpt catheter demonstrated stable position during deployment without slippage.

Table 8: QCA and Lesion Morphology Characteristics (Post-Procedure/In-Hospital

	Aggregate	Range	95% C.I.

Table 8: QCA and Lesion Morphology Characteristics (Post-Procedure/In-Hospital			
	Aggregate	**Range**	**95% C.I.**
Post ASC In-Lesion MLD	1.55 +/- 0.45 (190)	(0.51, 2.74)	(1.48, 1.61)
Post ASC In-Lesion %DS	43.48 +/- 14.66 (190)	(13.36, 79.84)	(41.38, 45.58)
Post ASC Dissection	13.6% (26/191)		(9.1%, 19.3%)
Type A	0.5% (1/191)		(0.0%, 2.9%)
Type B	6.3% (12/191)		(3.3%, 10.7%)
Type C	5.8% (11/191)		(2.9%, 10.1%)
Type D	1.0% (2/191)		(0.1%, 3.7%)
Type E	0.0% (0/191)		(0.0%, 1.9%)
Type F	0.0% (0/191)		(0.0%, 1.9%)
Final In-Lesion MLD	2.34 +/- 0.42 (219)	(1.31, 3.45)	(2.29, 2.40)
Final In-Lesion %DS	17.73 +/- 7.18 (219)	(3.65, 39.84)	(16.78, 18.69)
Final Thrombus	0.5% (1/219)		(0.0%, 2.5%)
Final Dissection	0.5% (1/218)		(0.0%, 2.5%)
Type A	0.0% (0/218)		(0.0%, 1.7%)
Type B	0.5% (1/218)		(0.0%, 2.5%)
Type C	0.0% (0/218)		(0.0%, 1.7%)
Type D	0.0% (0/218)		(0.0%, 1.7%)
Type E	0.0% (0/218)		(0.0%, 1.7%)
Type F	0.0% (0/218)		(0.0%, 1.7%)
Post ASC Acute Gain	0.77 +/- 0.43 (190)	(-0.28, 1.93)	(0.71, 0.83)
Final In-Lesion Acute Gain	1.57 +/- 0.41 (219)	(0.18, 2.73)	(1.51, 1.62)
Slippage	0.0% (0/206)		(0.0%, 1.8%)
Slippage - slight	0.0% (0/206)		(0.0%, 1.8%)
Slippage - moderate	0.0% (0/206)		(0.0%, 1.8%)
Slippage - severe	0.0% (0/206)		(0.0%, 1.8%)
Balloon to Artery Ratio	0.99 +/- 0.14 (201)	(0.67, 1.43)	(0.97, 1.00)
ASC = AngioSculpt, MLD = Minimal lumen diameter (mm), %DS = percent diameter stenosis (%) In a small number of cases, post-AngioSculpt angiograms were not performed and only the final angiograms were available. Balloon to artery ratio was determined by post-hoc core lab QCA evaluation.			

A summary of CEC adjudicated MACE is provided in **Table 9**. The cumulative MACE rate was calculated by including only one event per patient. Two patients experienced more than one MACE (Q wave MI with TLR-PTCA and Q wave MI with TLR-CABG). Thus, there were a total of seven MACE among five patients. The events included two (1%) non-Q wave MI, three (1.5%) Q wave MI, one (0.5%) TLR-PTCA and one (0.5%) TLR-CABG. "Confirmed" thrombosis was defined as angiographically documented thrombus or subacute closure within or adjacent to a previously successfully treated lesion at the time of clinically-driven (due to chest

pain and/or ECG changes) angiographic restudy for ischemia. Thrombosis is "presumed" in the absence of angiography.

Table 9: Cumulative MACE Rates (Through 14 Days Folow-Up)		
Cumulative Complications	**Aggregate**	**95% C.I.**
MACE (Death, MI, TLR)	2.5% (5/200)	(0.8%, 5.7%)
Death	0.0% (0/200)	(0.0%, 1.8%)
MI (Q or Non-Q)	2.5% (5/200)	(0.8%, 5.7%)
Q wave MI	1.5% (3/200)	(0.3%, 4.3%)
Non-Q wave MI	1.0% (2/200)	(0.1%, 3.6%)
TLR	1.0% (2/200)	(0.1%, 3.6%)
TLR PTCA	0.5% (1/200)	(0.0%, 2.8%)
TLR CABG	0.5% (1/200)	(0.0%, 2.8%)
Target Vessel Revascularization(TVR)	1.0% (2/200)	(0.1%, 3.6%)
TVR PTCA	0.5% (1/200)	(0.0%, 2.8%)
TVR CABG	0.5% (1/200)	(0.0%, 2.8%)
Thrombosis	1.5% (3/200)	(0.3%, 4.3%)
Confirmed	1.0% (2/200)	(0.1%, 3.6%)
Presumed	0.5% (1/200)	(0.0%, 2.8%)

Further analysis of the data for the 35 ISR lesions treated compared to the *de novo* lesions demonstrated no significant difference in primary efficacy and safety endpoints or in any of the MACE components between the two groups. The results are summarized in **Table 10.**

Table 10: Comparison of Outcomes in Patients with ISR vs. *de novo* Lesions				
Outcomes	**ISR**	***de novo***	**Combined**	**p-value**
Procedural Success	100.0% (33/33)	98.2% (164/167)	98.5% (197/200)	1.0000
Clinical Success	100.0% (33/33)	97.0% (162/167)	97.5% (195/200)	0.5932
In-Hospital MACE Free	100.0% (33/33)	98.2% (164/167)	98.5% (197/200)	1.0000
Out-of-Hospital MACE Free	100.0% (33/33)	98.8% (165/167)	99.0% (198/200)	1.0000
Death Free	100.0% (33/33)	100.0% (167/167)	100.0% (200/200)	N/A
MI Free	100.0% (33/33)	97.0% (162/167)	97.5% (195/200)	0.5932
Q-Wave MI Free	100.0% (33/33)	98.2% (164/167)	98.5% (197/200)	1.0000
Non Q-Wave MI Free	100.0% (33/33)	98.8% (165/167)	99.0% (198/200)	1.0000

Table 10: Comparison of Outcomes in Patients with ISR vs. *de novo* Lesions				
Outcomes	**ISR**	***de novo***	**Combined**	**p-value**
TLR Free	100.0% (33/33)	98.8% (165/167)	99.0% (198/200)	1.0000
TLR CABG Free	100.0% (33/33)	99.4% (166/167)	99.5% (199/200)	1.0000
TLR PTCA Free	100.0% (33/33)	99.4% (166/167)	99.5% (199/200)	1.0000

IVUS Sub-study:
IVUS was performed in a subset of 80 patients pre- and post-treatment with the AngioSculpt catheter to evaluate the morphologic effects of the device on the plaque and to further confirm device safety. Of these, 72 IVUS images were evaluable post-treatment, and showed an increase in minimal lumen area from 2.42 ± 0.98 mm^2 at baseline to 3.05 ± 1.04 mm^2 post-AngioSculpt treatment. There were no adventitial dissections or intramural hematomas, and there was no thrombus formation or any other evidence of unanticipated vessel injury following AngioSculpt treatment at the lesion site or in the adjacent reference vessel segments.

B. Outside of the United States (OUS) Clinical Investigations

Objective: To evaluate the safety and efficacy of the AngioSculpt catheter in a wide range of coronary artery lesions in both native vessels and following in-stent restenosis (ISR).

Conclusion: The initial OUS AngioSculpt clinical investigation shows favorable clinical results in a wide range of *de novo* lesions and in-stent restenosis (ISR). All patients completed 14-28 day post procedure clinical follow-up and experienced no major adverse cardiac events (MACE: death, Q wave or non-Q wave myocardial infarction, and target lesion revascularization) in-hospital or post discharge related to the study device.

Design: A multi-center clinical investigation was performed at two OUS sites, which enrolled a total of forty-five (45) patients.

Endpoints: The study was designed to evaluate the following primary performance and safety objectives:

- To demonstrate that the incidence and severity of device-related complications at one month follow-up are acceptably low.
- To demonstrate successful percutaneous revascularization (defined as a reduction in target lesion diameter stenosis to $\leq 50\%$ following completion of all interventions) when a stable (slip-resistant) platform for coronary artery dilatation is desired.

Eligibility Criteria: The study population included adult patients scheduled to undergo clinically indicated percutaneous coronary intervention at the designated clinical site for this study. All study candidates were required to provide, or have an authorized

legal representative provide, documented informed consent. Study subjects were required to understand the nature of the procedure, give informed consent, and be available for 14-28 day follow-up. Key patient selection criteria were:

- At least 18 years of age or older, or age of legal consent
- Planning to undergo clinically indicated percutaneous coronary intervention
- Target lesion reference vessel size 2.0 - 4.0 mm diameter
- Target lesion length $\leq$ 30 mm
- Target lesion severity $\geq$ 60% diameter stenosis

Methods: Following informed consent, patients referred for percutaneous coronary intervention (PCI) and meeting the study inclusion criteria underwent treatment with the AngioSculpt catheter. Enrolled patients had routine pre-procedure baseline assessments and were pre-medicated according to institutional protocol for percutaneous coronary interventions. At one investigational site, when technically possible, IVUS of the target lesion was performed prior to device deployment.

Coronary angiography of the target lesion was performed following completion of each device treatment and prior to adjunctive stent placement. IVUS of the target lesion was performed following device treatment (and prior to adjunctive stent placement). Additional clinically indicated interventions (e.g. stent placement) were performed and coronary angiography of the target lesion following completion of all interventions was performed in the original views. The AngioSculpt catheter was rated by the investigators after each use on pushability, trackability and passability on a scale of 0 - 4 (0=worst, 4=best) and on the degree of slippage during the procedure on a scale of 0 - 3 (0=none, 3=severe slippage).

Results of the evaluations and procedure assessments, including post-procedure adverse events and patient withdrawals, were documented on study-specific case report forms (CRFs).

One clinical investigator voluntarily arranged for eleven (11) ISR patients to return for follow-up examination six months after the interventional procedure. The investigator's assessments conducted at the 6-month follow-up included angiography and IVUS.

Results: The AngioSculpt catheter was utilized in 45 patients and a total of 46 lesions, consisting of 32 native vessels and 14 ISR lesions. The AngioSculpt catheter was successfully deployed in all 46 lesions. IVUS imaging demonstrated plaque scoring and luminal expansion following treatment with the AngioSculpt catheter with no perforations or other evidence of unanticipated vessel injury. There were no device-related major adverse cardiac events (MACE: death, Q wave or non-Q wave myocardial infarction, ischemia driven target lesion revascularization) encountered during hospitalization or during the follow-up period (24.8±8.5 days). There were no instances of device malfunction, retained device components or embolization. There was no evidence of any damage or deterioration in any of the devices. Overall, the

device was reported as being very deliverable. There was no clinically significant slippage associated with AngioSculpt use.

C. Gender Bias

The proportion of men and women enrolled in the U.S. pivotal study were consistent with previous studies of devices used in percutaneous interventions. Subgroup analysis was conducted to evaluate the effect of gender on primary efficacy outcomes. Gender was not significantly associated with the clinical outcomes. Since clinical outcomes were not associated with gender, these data demonstrated that gender was not an influencing factor on safety or effectiveness.

XI. CONCLUSIONS FROM CLINICAL AND NON-CLINICAL STUDIES

The safety and effectiveness of the AngioSculpt Scoring Balloon Catheter have been demonstrated by results obtained from biocompatibility testing, engineering bench testing, cadaver and *in vivo* animal testing, sterilization and shelf life testing, and clinical investigations.

The results of the OUS and US clinical investigations demonstrate that the AngioSculpt catheter is safe with an acceptably low rate of complications and effective in a broad range of coronary artery lesions consistent with its intended use. Together, the clinical and non-clinical test results provide reasonable assurance that the AngioSculpt catheter is safe and effective when used as indicated and in accordance with the instructions for use.

XII. PANEL RECOMMENDATION

In accordance with the provisions of section 515(c)(2) of the act as amended by the Safe Medical Devices Act of 1990, this PMA was not referred to the Circulatory Systems Panel, and FDA advisory committee, for review and recommendation because the information in the PMA substantially duplicates information previously reviewed by this panel.

XIII. CDRH DECISION

CDRH issued a letter to AngioScore, Inc. on March 30, 2006, advising that the PMA was approvable subject to acceptable results from inspections of the manufacturing facilities. The applicant's manufacturing facilities were inspected and found to be in compliance with the Quality System Regulation (21 CFR 820). CDRH approved this PMA application on January 8, 2007.

XIV. APPROVAL SPECIFICATIONS

Directions for Use: See the labeling.
Hazards to Health from Use of the Device: See Indications, Contraindications, Warnings, Precautions and Adverse Events in the labeling.
Postapproval Requirements and Restrictions: See approval order.